

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I SANDHIYA S (192521224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SANDHIYA S

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions.

Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions.

Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

ANSWERS

1. Given

- ❖ **File size:** 150 MB = **150,000,000 bytes**
- ❖ **Transport layer segment size:** 1500 bytes
- ❖ **Data Link layer overhead per frame:** 40 bytes
- ❖ **Bandwidth:** 50 Mbps = 50,000,000 bits/s

Number of Transport Layer Segments

$$\text{Number of segments} = \frac{150,000,000}{1500} = 100,000$$

Segments = 100,000

Number of Data Link Layer Frames

Each transport segment is encapsulated into **one frame**.

Frames transmitted = 100,000

Total Data Link Layer Overhead

Each frame adds **40 bytes** of header and trailer.

$$100,000 \times 40 = 4,000,000 \text{ bytes}$$

Total overhead = 4,000,000 bytes (4 MB)

Total Data Transmitted

$$150,000,000 + 4,000,000 = 154,000,000 \text{ bytes}$$

Convert to bits:

$$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$$

Transmission Time

$$\text{Time} = \frac{1,232,000,000}{50,000,000} = 24.64 \text{ seconds}$$

Estimated transmission time = 24.64 seconds (ignoring propagation delay)

Parameter	Value
File size	150 MB
Segment size	1500 bytes

Parameter	Value
Number of segments	100,000
Number of frames	100,000
Header/trailer per frame	40 bytes
Total overhead	4,000,000 bytes (4 MB)
Total transmitted data	154,000,000 bytes
Transmission time	24.64 seconds

How the Transport and Data Link Layers Ensure Reliable Communication

Transport Layer (e.g., TCP):

- ❖ Divides the file into manageable segments.
- ❖ Assigns **sequence numbers** so data can be reassembled correctly.
- ❖ Uses **acknowledgments (ACKs)** to confirm successful delivery.
- ❖ Retransmits lost or corrupted segments.
- ❖ Performs **flow control** to prevent overwhelming the receiver.
- ❖ Uses **error checking** (checksums) to detect data corruption.

Data Link Layer:

- Encapsulates each segment into a frame by adding headers and trailers.
- Uses **Frame Check Sequence (FCS)/CRC** to detect transmission errors.
- Ensures reliable delivery over a single network link.
- May retransmit corrupted frames (depending on the protocol).
- Controls access to the transmission medium using **MAC addressing** and media access mechanisms.

Together, the **Transport Layer** provides **end-to-end reliability** across the network, while the **Data Link Layer** ensures **reliable communication over each individual network link** by detecting and handling frame errors.

2. Given

- Sliding window size = 6 frames
- Data per frame = 1024 bytes
- Total frames to transmit = 48 frames
- One frame is lost in every window and must be retransmitted

Number of Transmission Windows Required

The number of windows is:

$$\text{Number of windows} = \frac{\text{Total frames}}{\text{Window size}} = \frac{48}{6} = 8$$

Number of transmission windows = 8

Total Number of Retransmissions

Since one frame is lost in every window, and there are 8 windows:

$$\text{Retransmissions} = 8 \times 1 = 8$$

Total retransmissions = 8 frames

Total Frames Sent Including Retransmissions

$$48 + 8 = 56 \text{ frames}$$

Total frames transmitted = 56 frames

Parameter	Value
Total frames	48
Window size	6 frames
Number of windows	8
Lost frames per window	1
Total retransmissions	8 frames
Total frames sent (including retransmissions)	56 frames
Data per frame	1024 bytes

How Flow Control Improves Communication Efficiency

Flow control is a mechanism used by the Transport Layer to regulate the rate of data transmission between the sender and receiver.

Benefits of Flow Control

1. Prevents Receiver Overload
 - Ensures the sender does not send data faster than the receiver can process or store it.
2. Reduces Packet Loss
 - Prevents buffer overflow at the receiver, reducing dropped frames and retransmissions.
3. Improves Network Utilization
 - Maintains a steady flow of data instead of frequent pauses and retransmissions.
4. Enhances Throughput
 - Allows multiple frames to be sent before waiting for acknowledgments, increasing efficiency.
5. Supports Reliable Communication
 - Works with acknowledgments and window management to ensure data is delivered correctly and in order.

Sliding Window Flow Control

With a window size of 6, the sender can transmit up to 6 frames before needing acknowledgments. As ACKs arrive, the window "slides" forward, allowing new frames to be sent. This keeps the communication channel busy and improves overall transmission efficiency compared to sending one frame at a time.

Conclusion: Flow control balances the transmission rate between sender and receiver, minimizes congestion and retransmissions, and increases the overall efficiency and reliability of data communication.

3. Given

- **Original IP packet size** = 12,000 bytes
- **MTU** = 1,500 bytes
- **IP header per fragment** = 20 bytes

Maximum Data per Fragment

Each fragment must fit within the MTU, including its IP header.

$$\text{Data per fragment} = 1500 - 20 = 1480 \text{ bytes}$$

Number of Fragments

Assuming the 12,000-byte packet includes the original header, the payload is:

$$12000 - 20 = 11980 \text{ bytes}$$

Number of fragments:

$$\lceil \frac{11980}{1480} \rceil = \lceil 8.0946 \rceil = 9$$

Total fragments created = 9

Total Header Overhead

Each fragment requires its own 20-byte IP header.

$$9 \times 20 = 180 \text{ bytes}$$

Total header overhead after fragmentation = 180 bytes

If we consider that the original packet already contained one 20-byte header, the **additional overhead introduced by fragmentation** is:

$$(9 - 1) \times 20 = 160 \text{ bytes}$$

Parameter	Value
Original packet size	12,000 bytes
MTU	1,500 bytes
IP header per fragment	20 bytes
Data carried per fragment	1,480 bytes
Number of fragments	9

Parameter	Value
Total fragment headers	180 bytes
Additional overhead due to fragmentation	160 bytes

How the Network Layer Ensures Successful Delivery of Fragmented Packets

The **Network Layer (IP)** handles fragmentation and reassembly so that packets can traverse networks with different MTU sizes.

1. Fragmentation

- When a packet is larger than the outgoing link's MTU, the router divides it into smaller fragments.
- Each fragment receives its own IP header.

2. Identification Field

- All fragments of the same original packet carry the same **Identification** value.
- This allows the destination to recognize which fragments belong together.

3. Fragment Offset

- Each fragment contains a **Fragment Offset** indicating its position within the original packet.
- The destination uses these offsets to place fragments in the correct order.

4. More Fragments (MF) Flag

- The MF flag indicates whether more fragments follow.
- The last fragment has MF = 0, signaling the end of the packet.

5. Routing

- Fragments are routed independently through the network.
- They may take different paths and arrive out of order.

6. Reassembly

- Only the destination host reassembles the fragments.

- Using the Identification field, Fragment Offset, and MF flag, it reconstructs the original 12,000-byte packet.

7. Error Handling

- If one or more fragments are lost, the complete packet cannot be reassembled.
- Higher-layer protocols such as TCP detect the loss and trigger retransmission.

Conclusion: The Network Layer enables large packets to traverse networks with smaller MTUs by fragmenting them into 9 fragments, adding 180 bytes of fragment headers (160 bytes of extra overhead due to fragmentation), and using identification, offsets, and flags to ensure correct reassembly at the destination.

4. Given Data

- Sliding window size = **6 frames**
- Data carried by each frame = **1024 bytes**
- Total frames transmitted = **48 frames**
- One frame is lost in every window and must be retransmitted

Number of Transmission Windows Required

The number of windows required is:

$$\text{Number of windows} = \frac{\text{Total frames}}{\text{Window size}} = \frac{48}{6} = 8$$

Number of transmission windows = 8

Total Number of Retransmissions

Since **one frame is lost in every window**, and there are **8 windows**:

$$\text{Total retransmissions} = 8 \times 1 = 8$$

Total retransmissions = 8 frames

Total Data Originally Sent

$$48 \times 1024 = 49,152 \text{ bytes}$$

Original user data transmitted = 49,152 bytes

Total Data Retransmitted

$$8 \times 1024 = 8,192 \text{ bytes}$$

Retransmitted data = 8,192 bytes

Parameter	Value
Total frames	48
Window size	6 frames
Number of windows	8
Lost frames per window	1
Total retransmissions	8 frames
Data per frame	1024 bytes
Original data transmitted	49,152 bytes
Retransmitted data	8,192 bytes

How Flow Control Improves Communication Efficiency

Flow control is a Transport Layer mechanism that regulates the amount of data a sender can transmit before receiving acknowledgments from the receiver.

Benefits of Flow Control

1. Prevents Receiver Buffer Overflow

- Ensures the sender does not send data faster than the receiver can process it.

2. Reduces Data Loss

- By avoiding congestion at the receiver, fewer frames are dropped and retransmitted.

3. Improves Throughput

- Multiple frames can be sent in one window before waiting for acknowledgments, making better use of available bandwidth.

4. Efficient Use of Network Resources

- Keeps the communication channel busy while minimizing idle time.

5. Supports Reliable Transmission

- Lost or corrupted frames can be detected and retransmitted without affecting the entire data transfer.

Sliding Window Example

With a **window size of 6**, the sender can transmit up to **6 frames** before requiring acknowledgments. As ACKs arrive, the window slides forward, allowing continuous transmission. This is much more efficient than a stop-and-wait approach, where only one frame is sent at a time.

Conclusion

- **Transmission windows required: 8**
- **Total retransmissions: 8 frames**
- Flow control improves communication efficiency by matching the sender's transmission rate to the receiver's capacity, reducing congestion, minimizing retransmissions, and increasing overall throughput.

5. Given

- **Video file size = 600 MB**
- **Checkpoint interval = 60 MB**
- **Failure occurs after = 390 MB transferred**

Number of Checkpoints Created Before Failure

The Session Layer inserts a checkpoint every 60 MB:

Data Transferred Checkpoint

60 MB	1
120 MB	2
180 MB	3
240 MB	4
300 MB	5
360 MB	6

The next checkpoint would be at 420 MB, but the failure occurs at 390 MB.

Therefore,

$$\text{Number of checkpoints before failure} = \frac{360}{60} = 6$$

Checkpoints created before failure = 6

Data to be Retransmitted After Recovery

After a failure, transmission resumes from the **last successful checkpoint**.

- Last checkpoint = **360 MB**
- Failure occurred at = **390 MB**

Data transmitted after the last checkpoint:

$$390 - 360 = 30 \text{ MB}$$

This portion must be retransmitted.

Data to be retransmitted = 30 MB

Parameter	Value
Total file size	600 MB
Checkpoint interval	60 MB
Failure point	390 MB

Parameter	Value
Last completed checkpoint	360 MB
Number of checkpoints created	6
Data to be retransmitted	30 MB

Importance of Synchronization in the Session Layer

The **Session Layer (Layer 5 of the OSI model)** manages communication sessions between applications and provides **synchronization points (checkpoints)** during long data transfers.

Benefits of Synchronization

1. Reduces Retransmission

- After a failure, communication resumes from the last checkpoint rather than restarting from the beginning.

2. Improves Reliability

- Ensures long file transfers, multimedia streams, and database transactions can recover efficiently from interruptions.

3. Saves Time and Bandwidth

- Only the data sent after the most recent checkpoint must be retransmitted.

4. Supports Error Recovery

- Checkpoints act as recovery markers that help restore communication sessions.

5. Enhances Performance

- Particularly useful for large files and long-running sessions where restarting the entire transfer would be costly.

Conclusion

For the 600 MB video transfer:

- **6 synchronization checkpoints** are created before the failure.

- Only **30 MB** of data must be retransmitted after recovery.
- Session Layer synchronization improves efficiency, reliability, and recovery by allowing communication to resume from the latest checkpoint instead of retransmitting the entire file.

6. Given

- **Original file size** = 900 MB
- **Compression** = 40%
- **Encryption overhead** = 5% of the compressed size

1. Compressed File Size

A 40% compression means the file size is reduced by 40%.

$$\text{Compressed size} = 900 \times (1 - 0.40) = 900 \times 0.60 = 540 \text{ MB}$$

Compressed file size = 540 MB

Encrypted File Size

Encryption adds 5% overhead to the compressed size.

$$\text{Encryption overhead} = 540 \times 0.05 = 27 \text{ MB} \quad \text{Encrypted file size} = 540 + 27 = 567$$

Encrypted file size = 567 MB

Total Percentage Reduction Compared to the Original File

Reduction in size:

$$900 - 567 = 333 \text{ MB}$$

Percentage reduction:

$$\frac{333}{900} \times 100 = 37\%$$

Total percentage reduction = 37%

Parameter	Value
Original file size	900 MB
Compression reduction	40%
Compressed file size	540 MB
Encryption overhead	27 MB
Encrypted file size	567 MB
Net size reduction	333 MB
Total percentage reduction	37%

How the Presentation Layer Improves Transmission Efficiency and Security

The **Presentation Layer (Layer 6 of the OSI model)** is responsible for data representation, compression, encryption, and translation between different data formats.

1. Improving Transmission Efficiency

- **Compression** reduces the amount of data that must be transmitted.
- Smaller files require:
 - Less bandwidth
 - Shorter transmission time
 - Lower storage requirements
- In this example, the file size is reduced from **900 MB to 540 MB** before transmission.

2. Improving Security

- **Encryption** converts readable data (plaintext) into an unreadable form (ciphertext).
- Protects data from unauthorized access during transmission.
- Only authorized users with the correct decryption key can recover the original data.

3. Data Format Translation

- Converts data between different character encoding and data formats.
- Ensures compatibility between heterogeneous systems.

4. Data Integrity and Privacy

- Encryption helps maintain confidentiality.
- Prevents attackers from understanding intercepted data.

Conclusion

- **Compressed file size: 540 MB**
- **Encrypted file size: 567 MB**
- **Overall reduction compared to the original file: 37%**

The Presentation Layer enhances **efficiency** through compression and **security** through encryption, enabling faster and safer data transmission across networks.

7. Given

- **Total data to transfer** = 3 GB = 3000 MB (using decimal units)
- **Request size** = 3 MB
- **Effective throughput** = 120 Mbps

Total Number of Requests Generated

$$\text{Number of requests} = \frac{\text{Total data}}{\text{Request size}} = \frac{3000 \text{ MB}}{3 \text{ MB}} = 1000$$

Total requests generated = 1000

Total Transmission Time

First, convert the total data size to bits:

$$3000 \text{ MB} \times 8 = 24000 \text{ Mb}$$

Using the throughput:

$$\text{Transmission Time} = \frac{\text{Total Data (Mb)}}{\text{Throughput (Mbps)}} = \frac{24000}{120} = 200 \text{ seconds}$$

Estimated transmission time = 200 seconds

(approximately **3 minutes 20 seconds**)

Parameter	Value
Total data size	3 GB (3000 MB)
Request size	3 MB
Number of requests	1000
Throughput	120 Mbps
Total data in megabits	24,000 Mb
Transmission time	200 seconds ($\approx 3 \text{ min } 20 \text{ s}$)

How the Application Layer Provides Reliable Services to Users

The **Application Layer (Layer 7 of the OSI model)** is the layer closest to the end user and provides network services directly to applications such as FTP, HTTP, SMTP, and DNS.

1. User-Friendly Network Access

- Provides interfaces that allow users and applications to access network resources without dealing with lower-layer details.

2. File Transfer Services

- Protocols such as **FTP** manage file uploads and downloads.
- Large files are divided into manageable requests or data blocks for efficient transfer.

3. Error Reporting and Recovery

- Detects transfer failures and informs the user.
- Supports restarting interrupted transfers and retransmitting missing data.

4. Authentication and Authorization

- Verifies user credentials before granting access to files and services.
- Protects network resources from unauthorized users.

5. Data Integrity

- Works with lower layers to ensure files arrive completely and correctly.
- FTP can verify file size and transfer status to confirm successful delivery.

6. Service Management

- Establishes, maintains, and terminates application sessions.
- Provides status information, acknowledgments, and progress monitoring.

Conclusion

For the FTP transfer:

- **Number of requests generated: 1000**
- **Estimated transmission time: 200 seconds (3 minutes 20 seconds)**

The **Application Layer** ensures reliable service by providing file transfer management, user authentication, error handling, session control, and an easy-to-use interface that allows users to transfer data efficiently and securely.

8.

Given

Layer	Processing Delay
Application Layer	4 ms
Transport Layer	3 ms
Network Layer	5 ms
Data Link Layer	2 ms
Physical Layer	1 ms

Additional delays:

- **Propagation delay** = 18 ms
- **Transmission delay** = 12 ms

Total Processing Delay

$$\text{Processing Delay} = 4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total End-to-End Delay

$$\text{Total Delay} = \text{Processing Delay} + \text{Transmission Delay} + \text{Propagation Delay} = 15 + 12 + 18 = 45 \text{ ms}$$

Total End-to-End Delay = 45 ms

Delay Type	Value
Application Layer delay	4 ms
Transport Layer delay	3 ms
Network Layer delay	5 ms
Data Link Layer delay	2 ms
Physical Layer delay	1 ms
Total Processing Delay	15 ms
Transmission Delay	12 ms
Propagation Delay	18 ms
Total End-to-End Delay	45 ms

Contribution of Each OSI Layer to Communication

1. Application Layer (4 ms)

- Provides network services directly to user applications.
- Supports protocols such as HTTP, FTP, SMTP, and DNS.
- Generates and receives user data.

2. Transport Layer (3 ms)

- Segments data into smaller units.
- Provides end-to-end communication.
- Handles flow control, error recovery, acknowledgments, and retransmissions (e.g., TCP).

3. Network Layer (5 ms)

- Determines the best route from source to destination.
- Performs logical addressing using IP addresses.
- Handles routing and packet forwarding.

4. Data Link Layer (2 ms)

- Encapsulates packets into frames.
- Uses MAC addresses for local delivery.
- Detects transmission errors using CRC/FCS.

5. Physical Layer (1 ms)

- Converts bits into electrical, optical, or radio signals.
- Handles the actual transmission medium and signal encoding.

Other Delays

Transmission Delay (12 ms)

- Time required to place all packet bits onto the communication link.

$$\text{Transmission Delay} = \frac{\text{Packet Size}}{\text{Bandwidth}}$$

Propagation Delay (18 ms)

- Time taken for the signal to travel through the medium from sender to receiver.

$$\text{Propagation Delay} = \frac{\text{Distance}}{\text{Propagation Speed}}$$

Conclusion

The packet experiences:

$$\text{Total End-to-End Delay} = 45 \text{ ms}$$

This delay consists of:

- **15 ms** of processing across the OSI layers,
- **12 ms** of transmission delay, and
- **18 ms** of propagation delay.

Each OSI layer performs a specific function—ranging from application services to physical signal transmission—working together to ensure reliable and efficient communication.

9. Given

- **Useful data** = 80 MB = 80,000,000 bytes
- **Frame size** = 1600 bytes
- **Protocol overhead per frame** = 80 bytes

Therefore,

$$\text{Useful data per frame} = 1600 - 80 = 1520 \text{ bytes}$$

Total Number of Frames Required

$$\text{Number of frames} = \left\lceil \frac{80,000,000}{1520} \right\rceil = \lceil 631.58 \rceil = 52,632$$

Total frames required = 52,632

Total Overhead Transmitted

$$\text{Total overhead} = 52,632 \times 80 = 4,210,560 \text{ bytes}$$

Total overhead transmitted = 4,210,560 bytes $\approx$ 4.21 MB

Transmission Efficiency

Transmission efficiency is the percentage of useful data in each frame.

$$\text{Efficiency} = \frac{\text{Useful Data}}{\text{Frame Size}} \times 100 = \frac{1520}{1600} \times 100 = 95\%$$

Transmission efficiency = 95%

(Equivalently, overhead percentage = $\frac{80}{1600} \times 100 = 5\%$.)

Parameter	Value
Useful data	80 MB
Frame size	1600 bytes
Overhead per frame	80 bytes
Useful data per frame	1520 bytes
Number of frames required	52,632
Total overhead transmitted	4,210,560 bytes (≈ 4.21 MB)
Transmission efficiency	95%
Overhead percentage	5%

How Protocol Overhead Affects Network Performance

Protocol overhead consists of headers, trailers, checksums, addresses, sequence numbers, and other control information added by various OSI layers.

Positive Effects

- **Reliability:** Enables error detection and recovery.
- **Routing:** Provides source and destination addressing.
- **Flow Control:** Prevents congestion and receiver overload.
- **Security:** Supports authentication and encryption.
- **Session Management:** Maintains communication state.

Negative Effects

- **Consumes Bandwidth:** Overhead occupies part of every frame, leaving less room for user data.
- **Increases Transmission Time:** More bits must be transmitted.

- **Reduces Throughput:** Effective data transfer rate becomes lower than the raw link speed.
- **Adds Processing Load:** Devices must generate, read, and process protocol headers.

Conclusion

For the transfer of **80 MB** of useful data:

- **Frames required: 52,632**
- **Total overhead transmitted: 4,210,560 bytes (≈ 4.21 MB)**
- **Transmission efficiency: 95%**

Protocol overhead is essential for reliability, addressing, and control, but it also consumes bandwidth and slightly reduces overall network efficiency.

10. Given

- **Original medical record size** = 240 MB = 240,000,000 bytes
- **Compression by Presentation Layer** = 30%
- **Transport Layer segment size** = 1400 bytes
- **Data Link Layer overhead per frame** = 60 bytes

Compressed File Size

A 30% compression reduces the file size to 70% of the original.

Compressed Size = 240,000,000 $\times$ 0.70 = 168,000,000 bytes = 168 MB

Compressed file size = 168 MB

Total Number of Segments Transmitted

$$\text{Number of Segments} = \frac{168,000,000}{1400} = 120,000$$

Since it divides exactly:

Total segments transmitted = 120,000

Assuming one segment is encapsulated into one frame:

Total frames transmitted = 120,000

Total Protocol Overhead Introduced

Each frame adds **60 bytes** of overhead.

$$\text{Total Overhead} = 120,000 \times 60 = 7,200,000 \text{ bytes} = 7.2 \text{ MB}$$

Total protocol overhead = 7,200,000 bytes (7.2 MB)

Final Amount of Data Transmitted Across the Physical Layer

$$\text{Final Data} = \text{Compressed Data} + \text{Overhead} = 168,000,000 + 7,200,000 = 175,200,000 \text{ bytes}$$

Final data transmitted across the Physical Layer = 175.2 MB

Parameter	Value
Original file size	240 MB
Compression	30%
Compressed file size	168 MB
Segment size	1400 bytes
Number of segments	120,000
Data Link overhead/frame	60 bytes
Total overhead	7,200,000 bytes (7.2 MB)
Final transmitted data	175,200,000 bytes (175.2 MB)

How Multiple OSI Layers Work Together for Secure and Reliable Communication

1. Presentation Layer (Layer 6)

- Compresses the medical record to reduce bandwidth usage and transmission time.

- May encrypt sensitive patient information to ensure confidentiality and privacy.

2. Transport Layer (Layer 4)

- Divides the compressed data into 1400-byte segments.
- Provides end-to-end reliability using sequence numbers, acknowledgments, error recovery, and flow control.
- Ensures all segments arrive correctly and in order.

3. Network Layer (Layer 3)

- Assigns logical addresses (IP addresses).
- Selects the best route through the healthcare network.
- Delivers packets from source to destination across multiple networks.

4. Data Link Layer (Layer 2)

- Encapsulates segments into frames.
- Adds 60-byte headers/trailers for addressing and error detection.
- Uses CRC/FCS to detect transmission errors on each link.

5. Physical Layer (Layer 1)

- Converts frames into electrical, optical, or wireless signals.
- Transmits the final 175.2 MB of data through the communication medium.

Conclusion

For the 240 MB medical record:

- **Compressed file size: 168 MB**
- **Segments transmitted: 120,000**
- **Protocol overhead introduced: 7.2 MB**
- **Final data transmitted across the Physical Layer: 175.2 MB**

The OSI layers collaborate to provide efficient, secure, and reliable communication, which is essential for transmitting critical healthcare information in IoT-enabled medical networks.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand